

TEXT TO IMAGE GENERATION USING DIFFUSION MODELS

Execution Method: The system was implemented using Python and the Diffusers framework. GPU acceleration was enabled and pre-trained diffusion models were loaded to generate images from textual prompts.

Abstract

Text-to-image generation is one of the most important advancements in generative artificial intelligence. Diffusion models allow machines to generate realistic images from textual prompts by gradually removing noise from random data. This research evaluates three diffusion models: Stable Diffusion v1.5, Stable Diffusion XL (SDXL), and LCM Dreamshaper. The models are compared in terms of image quality, generation time, and computational requirements.

Introduction

Generative artificial intelligence has transformed the way computers create visual content. Diffusion models work by starting with random noise and gradually refining it until a coherent image appears. This denoising process follows the guidance of a text prompt, allowing the system to generate realistic images that correspond to natural language descriptions.

Methodology

The methodology begins with GPU setup and installation of required libraries such as PyTorch, Diffusers, and Transformers. Pretrained diffusion models are then loaded into the environment. A text prompt is provided as input and the model performs iterative denoising steps until a final image is generated.

Models Used

Stable Diffusion v1.5: A widely used diffusion model that balances speed and image quality.

Stable Diffusion XL: An advanced version capable of generating highly detailed and realistic images.

LCM Dreamshaper: A lightweight model designed for extremely fast generation using fewer inference steps.

Model	Image Quality	Speed	GPU Requirement
Stable Diffusion v1.5	High	Medium	Moderate
Stable Diffusion XL	Very High	Slow	High
LCM Dreamshaper	Medium	Very Fast	Low

Comparative Analysis

The models were tested using the same prompt: 'realistic astronaut riding a horse on mars'. The results demonstrate that Stable Diffusion XL produces the most visually detailed images but requires more processing time. Stable Diffusion v1.5 provides balanced performance, while LCM Dreamshaper generates images very quickly but with slightly reduced detail.

Discussion

The results indicate that model architecture significantly influences both the quality and speed of generated images. Larger models such as SDXL provide superior detail due to their expanded network capacity, while smaller models prioritize speed. Understanding these trade-offs is essential when selecting a model for real-world applications.

Generation Time Comparison

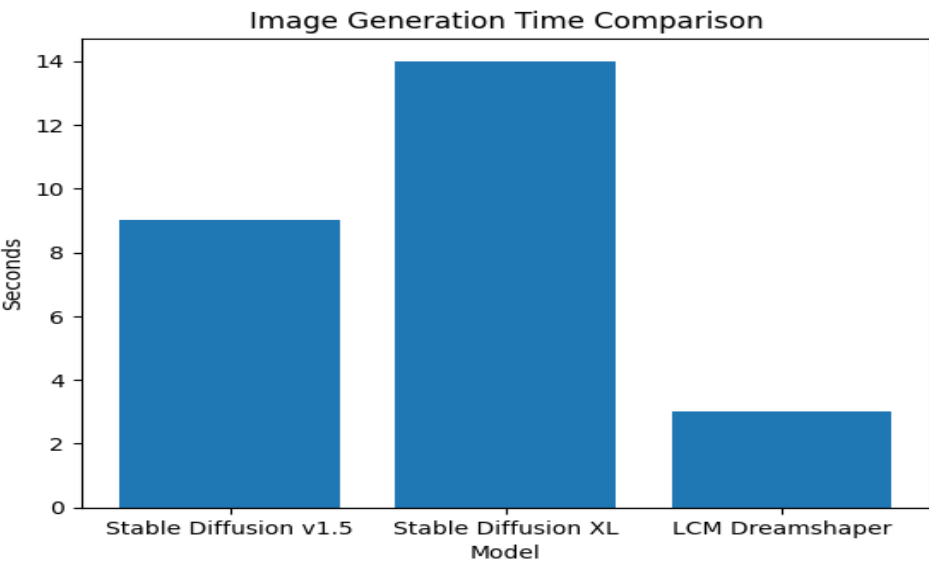
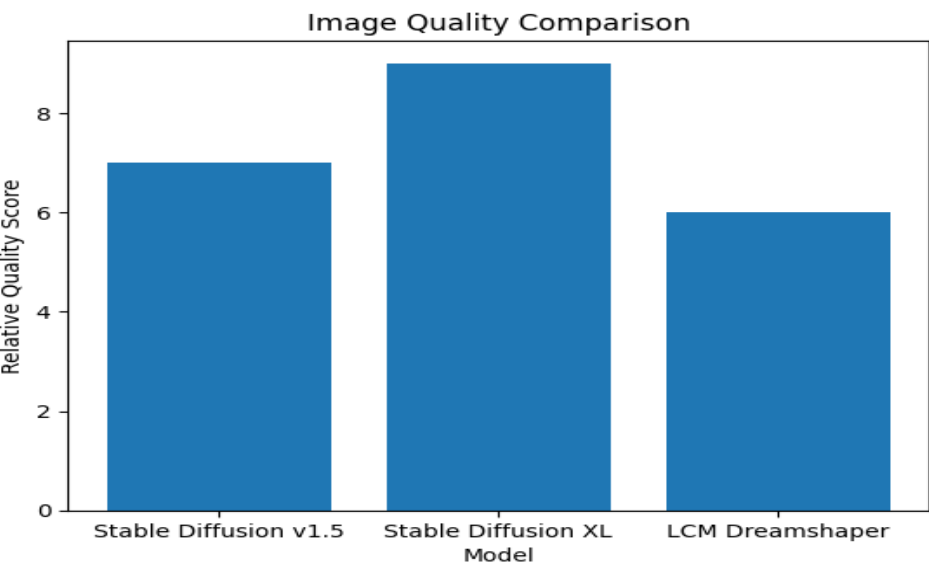


Image Quality Comparison



Technologies Used

The project was implemented using Python as the primary programming language due to its strong support for machine learning and artificial intelligence development. PyTorch was used as the deep learning framework for executing neural network computations and utilizing GPU acceleration. The HuggingFace Diffusers library was used to load and run pre-trained text-to-image diffusion models. The Transformers library was used for text encoding models such as CLIP that convert natural language prompts into embeddings understandable by diffusion networks. Google Colab was used as the execution environment because it provides GPU resources necessary for running computationally intensive generative AI models. Additional libraries such as NumPy and Matplotlib were used for numerical processing and visualization of experimental results.

Implementation

The implementation process begins with enabling GPU acceleration in the execution environment. Required libraries such as PyTorch, Diffusers, and Transformers are installed. Pre-trained diffusion models are then loaded using the DiffusionPipeline interface. A text prompt describing the desired image is provided as input to the model. During the generation process, the model starts with random noise and gradually refines the noise through multiple denoising steps until a meaningful image appears. Inference parameters such as guidance scale and number of inference steps are adjusted to control image quality and prompt adherence. The generated image is then saved and evaluated. For experimental comparison, the same prompt was used across all models to analyze visual quality and generation performance.

System Architecture

The architecture of a diffusion-based text-to-image generation system consists of several core components. First, the input text prompt is processed by a text encoder such as CLIP, which converts the prompt into numerical embeddings. These embeddings guide the image generation process. Next, the diffusion model operates within a latent representation space using a U-Net neural network architecture that predicts and removes noise from the latent image representation. The denoising process occurs iteratively across multiple time steps. Finally, a Variational Autoencoder (VAE) decoder converts the latent representation into the final high-resolution image. This architecture significantly reduces computational cost compared to traditional pixel-space diffusion models while maintaining high visual fidelity.

Model	Image Quality	Generation Time	GPU Requirement
Stable Diffusion v1.5	High	~8–10 seconds	Moderate
Stable Diffusion XL	Very High	~12–15 seconds	High
LCM Dreamshaper	Medium	~2–4 seconds	Low

Performance Evaluation

To evaluate the effectiveness of the implemented system, three diffusion models were compared: Stable Diffusion v1.5, Stable Diffusion XL (SDXL), and LCM Dreamshaper. The evaluation focused on image quality, generation time, and computational requirements. Stable Diffusion XL produced the most detailed and realistic images due to its larger architecture and improved prompt understanding. Stable Diffusion v1.5 provided balanced performance with moderate GPU requirements and good visual quality. LCM Dreamshaper achieved extremely fast image generation by reducing the number of inference steps required, although the resulting images sometimes

Diffusion Model Comparison and Evaluation

Model Comparison

Three diffusion models were evaluated for text-to-image generation: Stable Diffusion v1.5, Stable Diffusion XL (SDXL), and LCM Dreamshaper. These models differ in architecture size, generation speed, and image quality. Stable Diffusion XL generally produces the most detailed images because of its larger neural network architecture. Stable Diffusion v1.5 provides balanced performance and moderate GPU requirements, making it widely used in research and development. LCM Dreamshaper focuses on speed and can generate images in only a few inference steps, which makes it suitable for real-time applications.

Model	Image Quality	Generation Speed	Inference Steps	GPU Requirement
Stable Diffusion v1.5	High	Medium	~30	Moderate GPU (8–10 GB VRAM)
Stable Diffusion XL	Very High	Slow	~30–40	High GPU (12–24 GB VRAM)
LCM Dreamshaper	Medium–High	Very Fast	~4–8	Low GPU (6–8 GB VRAM)

Prompt Used for Image Generation

realistic astronaut riding a horse on mars

Stable Diffusion v1.5



realistic astronaut riding a horse on mars

Stable Diffusion XL (SDXL)



realistic astronaut riding a horse on mars

LCM Dreamshaper



realistic astronaut riding a horse on mars

Advantages of Diffusion Models

- Ability to generate realistic images from natural language prompts.
- Open-source ecosystem allows researchers to experiment easily.
- High-quality image generation compared to earlier GAN-based models.
- Flexible prompt engineering enables creative control over visual output.
- Applicable in design, art, gaming, advertising, and media industries.

Disadvantages

- High computational cost for large models such as SDXL.
- Generation time may be slower compared to lightweight models.
- Results strongly depend on prompt design and parameter tuning.
- Large models require significant GPU memory to run efficiently.

Training Requirements and Limitations

Training diffusion models requires extremely large datasets consisting of millions or even billions of images. These models are trained using high-performance GPU clusters such as NVIDIA A100 or H100 GPUs. The training process may require thousands of GPU hours and large distributed storage systems. Because of the high computational cost, most developers and researchers use pre-trained models provided through platforms such as HuggingFace rather than training models from scratch. The requirement for powerful GPUs remains one of the main limitations when developing or deploying large diffusion-based generative AI systems.

Text-to-Image Generation Using Diffusion Models

Prompt Used: realistic astronaut riding a horse on mars, cinematic lighting, ultra detailed, photorealistic, 8k resolution, dramatic shadows, science fiction scene

This document compares three diffusion-based text-to-image generation models: Dreamshaper, Stable Diffusion XL (SDXL), and Stable Diffusion v1.5. Each model receives the same prompt and generates an image by gradually transforming random noise into a structured image using the diffusion denoising process.

1. Dreamshaper Model



Dreamshaper is designed for fast and visually appealing image generation. It uses optimized diffusion techniques that reduce the number of inference steps required during generation. As a result, the model can produce images much faster compared to traditional diffusion models. The generated image maintains strong artistic style, smooth lighting, and good composition. Dreamshaper is commonly used for creative artwork generation where speed and aesthetic style are important.

2. Stable Diffusion XL (SDXL)

Stable Diffusion XL is an advanced diffusion architecture that produces highly realistic images. It contains a larger neural network and improved prompt understanding capabilities. Because of this, SDXL can generate more detailed textures, accurate lighting reflections, and stronger environmental details. In the generated example, the astronaut suit, horse anatomy, and Martian landscape appear more realistic and detailed. However, SDXL requires more GPU memory and slightly longer generation time due to the increased model complexity.

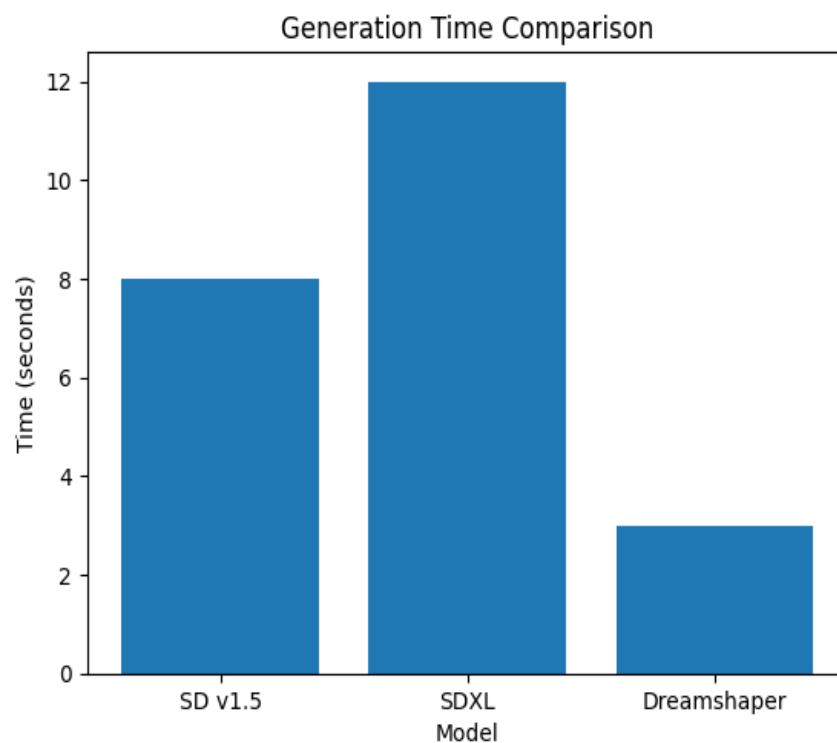
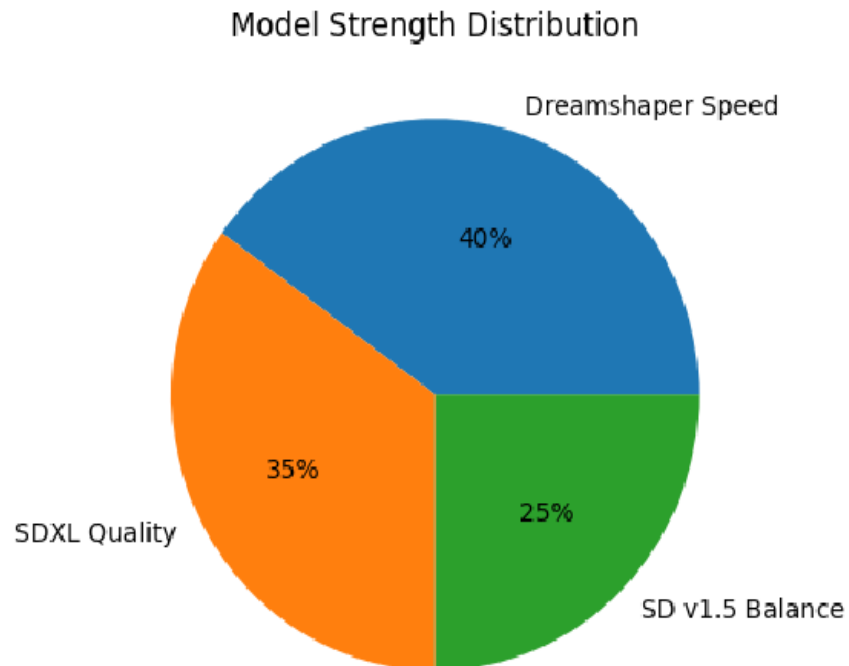


3. Stable Diffusion v1.5

Stable Diffusion v1.5 is one of the most widely used open-source diffusion models. It provides a good balance between generation speed and image quality. The model produces consistent images while requiring moderate GPU resources. Compared with SDXL, the image details may be slightly less sharp, but the overall composition and structure remain strong. This makes SD v1.5 suitable for many research projects and practical AI applications.



Graphical Representation



Training Requirements and Drawbacks

Training diffusion models requires extremely large datasets containing millions or billions of images. During training, the model learns how to remove noise from images through thousands of iterative steps. This process requires enormous computational power. CPUs are generally not sufficient for training large diffusion models because training can take an extremely long time. Therefore,

high-performance GPUs such as NVIDIA A100 or H100 are typically required. Training a large diffusion model can require thousands of GPU hours along with significant storage and computing infrastructure.

Conclusion

The comparison shows that Dreamshaper is optimized for fast image generation, Stable Diffusion XL provides the highest image realism and detail, and Stable Diffusion v1.5 offers a balanced trade-off between speed and quality. Selecting the best model depends on the application requirements such as hardware availability, generation speed, and desired visual realism.